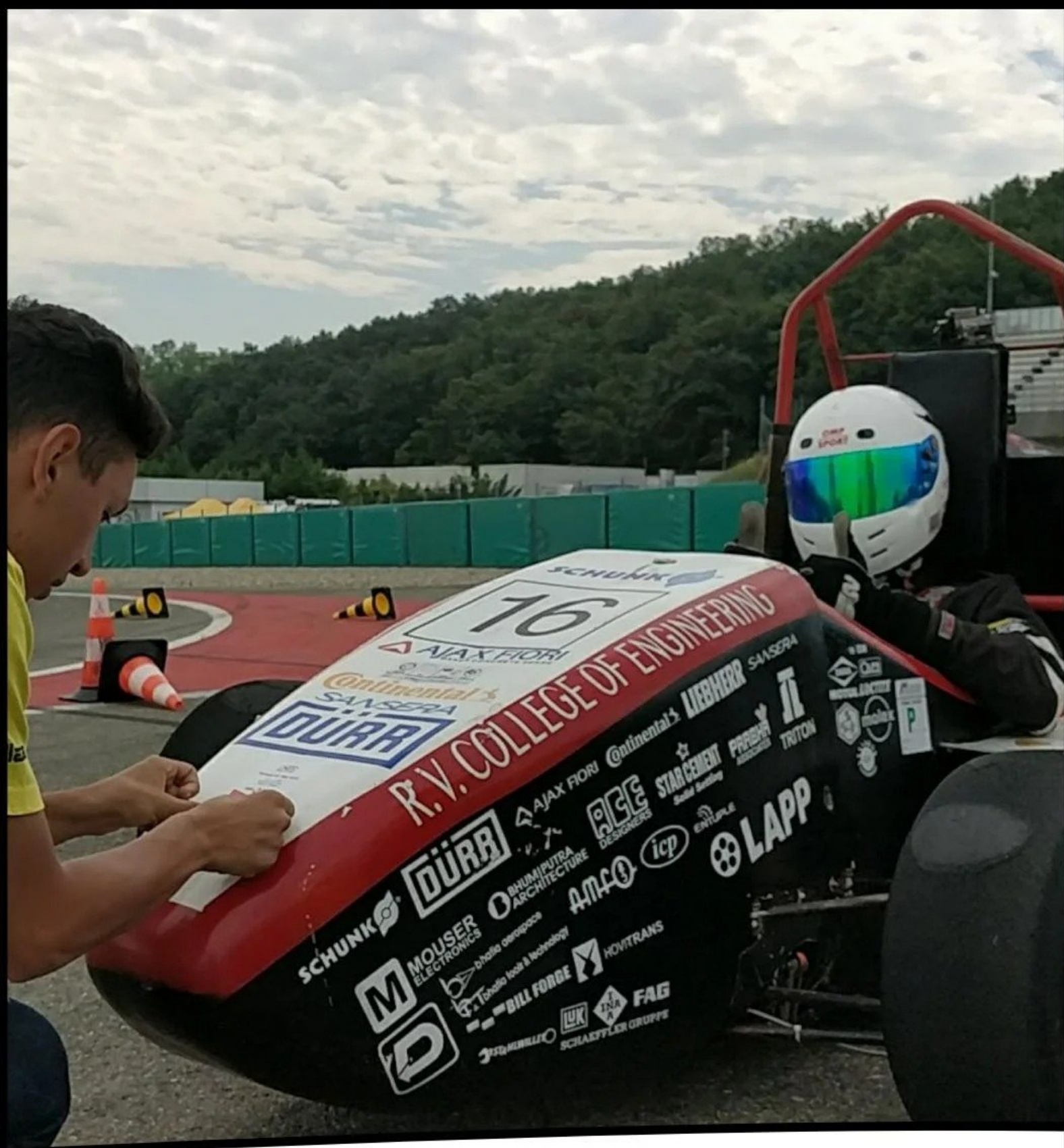




बड़हवा RACING

NEWSLETTER - August 2018



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What's Happening @ the Workshop?



Start of July, 2018 marked the beginning of the Pre-Design phase for the RZ-X9C and RZ-XXH while the RZ-X9H entered its manufacturing phase by the end of the month. RZ-X9C is expected to compete in the Formula Bharat as well as FS Italy event and RZ-X9H is expected to compete in the Formula Hybrid event, 2019

Since Ashwa Racing believes in excellence and innovation, it has implemented about 3-5% changes in its hybrid prototype, with the differential mount positions being changed, offering more strength and rigidity to the driveline, thereby improving the overall efficiency of the prototype. The vital bearings and fasteners have been received while the procurement of crucial OEM's such as the batteries and motors is in progress. The team is confident of starting the assembly phase at the earliest.



Inauguration Day

The Inauguration day was held on 29th July, 2018 in the college premises, to welcome freshers who would begin a new journey in the field of engineering. The team displayed two of its prototypes namely, X2 and X7H during the inauguration, briefly explaining about the project and the perks of being a part of it. The team interacted with students from various disciplines as well as their parents who showed keen interest in the project. We are thankful to the college authorities for letting us have an excellent opportunity to interact with enthusiastic students, their parents and industry professionals.

Formula Student Italy

With much higher hopes and preparations after Formula Bharat, Ashwa Racing took part in this year's Formula SAE Italy Event competing against the top combustion teams from across the globe. The event was held from 11th July 2018 to 15th July 2018.



Varano de Melegari, Province of Parma, Italy.

The X8C, which previously set the national record for acceleration at Formula Bharat 2018, had some minor changes made to it, for the Italy event. The prototype was powered by a GSX-R600cc engine.

The global objective of this event is to evaluate whether the prototypes built by students are worthy of being mass produced. This puts to test, not only engineering skills, but also the team's project management capabilities. This evaluation is done at many levels, which we know as the sub events within the competition.



Formula Student Italy

TECHNICAL SCRUTINY

Safety, good engineering practices, and alignment with environmental regulations (such as noise levels) are important criteria. With this in mind, the teams are expected to follow guidelines that are elaborated in a rulebook. Compliance with the guidelines are checked during technical inspection. This is followed by tilt, brake and noise test, clearing which, the teams are allowed to compete in dynamic events.

STATIC EVENTS

The static events comprised of the Design Event, Business Event and the Cost Event. The team gave their best in all the three events.

Design event is an assessment of engineering skills employed to make the prototype. Judges evaluate how well does the procedure of building the car align with the team's performance goals despite constraints. The objective of the Business Event is to assess the marketability and commercial viability of the prototype as a business proposition to venture capitalists/potential investors. The Cost Event is where the prototype is assessed for its value for money.



Formula Student Italy

DYNAMIC EVENTS

After clearing the static events, the team progressed to Dynamic Events that consisted of Acceleration, Skidpad, Autocross and Endurance event.

Skidpad Event: The track is in the shape of the number '8' and the goal of this event is to test the lateral motion of the car. The ability of the driver to maneuver the car through the path, without knocking out the cones is also a challenge. Our final timing for the skidpad event was 8.23s

Acceleration Event: Acceleration event tests the vehicle's longitudinal motion performance. The team stepped up to the expectations and performed excellently in the event, with an impressive timing of 4.78s.

However the Endurance event was put on hold indefinitely after the car spun out of control during the 8th lap and incurred a few damages. The team captain, Sweekruth Shetty, was behind the wheel and is safe by all means, after the crash.



MASERATI



Formula Student Italy

One of the most important phases of this entire process is Testing. The prototype was rigorously tested after FB 2018, the car ran over 22km during the testing sessions. The Business Presentations have improved over the years and X8C even bagged the second position for BP at Formula Bharat, 2018.

An important take away, from this Race Season, is to focus more on our driver training sessions and prepare ourselves for unforeseen circumstances, such as, crashes. The junior batches are now working towards an essential yet important goal : finishing Endurance.

The Italy event marked the end of another Race Season. Completion of three full events, with a top 10 finish in Formula Bharat and a Podium finish in Formula Hybrid, it has been a very fruitful event for Ashwa Racing. An important take away from this race season is the importance of ensuring that the drivers are adequately experienced. This coupled with logging in more testing time will improve our results at the competition. With all the ups and downs and slight mistakes that happen to slip our minds, it was an excellent Race season that will go down in Ashwa's steady legacy, mounting an onus the following batches to come to learn and atone from the mistakes made during the event and improve the overall performance of the team.



Driver Testing & Safety



The endurance event in this year's FS Italy was put on hold as the car spun out of control during the 8th lap and incurred a few damages. Although, the team captain Sweekruth Shetty was behind the wheel and was safe after the clash by virtue of the rigorous driver safety and testing practices at Ashwa Racing. Driver selection at Ashwa Racing comprises of many layers of vigorous testing.

Driver Safety and Testing comprises of a 5 point safety harness: a seatbelt that comprises of 5 belts. The driver suit is completely fireproof and covers the driver, top to bottom. The firewall is an aluminium sheet that separates the engine from the driver cockpit. Even while testing the car in college, safety measures are at a high and never taken for granted, with roads being blocked by cones and people standing at a safe distance from the periphery of the track.

At the event, there are people standing next to the track equipped with fire extinguishers at all times. Countermeasures are taken at all times to minimize damage and injuries. A safety officer present at the time of testing has a safety checklist and only when the car clears the safety check it is allowed to run. This drill is practised during testing as well. In conclusion, although there have been incidents of losing control over the car, we have reached a stage where the driver has never sustained

AUDIT - July 2018



R. V. COLLEGE OF ENGINEERING

अश्वरacing RACING

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R. V. COLLEGE OF ENGINEERING **अश्वरacing** RACING

Accounts for the period from 01-07-2018 to 31-07-2018

RECIEPTS	AMOUNT
Opening Balance in ING Vysya Bank RVCE Branch	
SB A/c 136010046665	₹ 3,25,234.37
SB A/c 136010058514	₹ 1,03,183.41
SB A/c 136010060793	₹ 15,69,200.43
seed money	₹ 96,840.00
sponsorship	₹ 3,31,000.00
Other deposits	₹ 44,256.00
TOTAL	₹ 24,69,714.21

PAYMENTS BY	AMOUNT
Chassis & Suspension	₹ 15,234.00
Powertrain & Smp	₹ 63,612.00
Electrical	₹ 48,602.00
Brakes & Logistics	₹ 1,35,401.50
Miscellaneous	₹ 7,950.00
SMS & Wire Charges	₹ 53.10
Cash In Hand	₹ 573.00
Closing Balance in ING Vysya Bank RVCE Branch	
SB A/c 136010046665	₹ 1,90,281.87
SB A/c 136010058514	₹ 1,07,806.31
SB A/c 136010060793	₹ 19,00,200.43
TOTAL	₹ 24,69,714.21

Checked and Found Correct.

Shreyash

Shreyash S Kalal
Finance Team

अश्वरacing
R.V. College of engineering

ALUMNUS OF THE MONTH



Eshwar has been one of the strong pillars of the 2016 Combustion team. Fondly called "God" by his very team mates, Eshwar has worked on the Steering & Finance module.

Eshwar is the very first name that comes to our minds when we look at the chassis getting welded. This not just states his exemplary skills on welding, but also his expertise in the manufacturing processes. He even held training sessions teaching his juniors on how to have a strong weld across various profiles of joints.

He is an encyclopedia, with thorough knowledge of every single operation that needs to be done on any component. Not just the operation, but also the right vendor to be approached for the same.

It wouldn't be wrong to state that he can prepare the operations sheet for any component which comes on the cost report. This expertise in the domain of operations and costing of components also made him get selected as one of the cost judges for the Formula Bharat event.

He has also played a key role in the selection of the UV joints for the 2018 prototypes: if we are at ease regarding clearing the Steering play rule, then we must accord due credits to Eshwar.

Having such strong technical foundations, Eshwar is by far one of the most down to earth members we would have ever had. He is always approachable for any support. Quite a few juniors try pulling his leg, but he still remains as friendly as one could get and this makes him more close to all his juniors.

Eshwar's broken Hindi, especially while conversing with his teammates, is still remembered and is something which gets us a smile on our face even today.

Thank you so much Eshu for guiding us and helping us not just design well, but also manufacture the components with an equal level of prowess.

What one must always remember about Eshwar is that...

ESHU ROCKS..9!!

SPONSOR FOCUS



Created in 1863, Solvay is a global company driven by proud and committed chemists. With their historical anchorage in Europe, their products serve diversified markets worldwide, from consumer goods to energy, with one main aim - to improve quality of life and customer performance. The company ensures long-term sustainable growth for its customers and the planet, thereby focusin its Research and innovation in the fields of Advanced Materials, Renewable Chemistry, Sustainable Energy Solutions & Consumer Chemicals. The team extends its heartfelt gratitude to Solvay for their continued financial support for the 2018-19 race season.



Ace Designers, led by three experienced design engineers and powered by a dynamic team of more than 500 members is one of India's largest machine tool manufacturing companies. Ace has rapidly grown to become India's largest manufacturer of CNC turning centres and has remained the undisputed leader for nearly two decades. With a strategic focus on manufacturing of CNC turning centres and has developed a wide range of indigenous cost effective products to meet the changing needs of its customers. Ace has ensured product excellence through quality driven manufacturing processes supported by meticulously planned world class infrastructure. The team wholeheartedly thanks Ace Designers for their continued financial support for the upcoming race season.



Lakshmi Precision Screws Ltd. is a high tensile manufacturing unit which was established in the year 1972. The Company is a pioneer in providing fastening technology globally having Joint Ventures, License Agreements, and Alliances with different fastening companies globally and catering the needs of various sectors such as Wind Energy, Oil & Gas, Locomotives, Automobiles, Agriculture Equipments (Tractors), Machine Building and different Industrial and its sub-con items. The team thanks LPS for their continued support for the 2018-19 race season.

Video Testimonial



Mr. Dilbagh Gill
Team Principal of Mahindra
Formula E Racing



Mr. Hubert Reilard,
Hon'ble ex-President of
Indo German Chamber of
Commerce and ex-
Managing Director of
,EFD Induction



Mr. Rajat Kabra
Alumni, Ashwa Racing
Team Captain of Batch
of 2016



Mr. Mayur Bhagat
Director
Tuflite Polymers



Ms. Diksha Bhagat
Alumni, Ashwa Racing Batch of 2016
Presently working in Toyota Kirloskar Motors

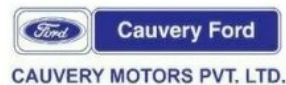
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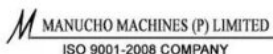


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HI-FLEX TECHNOLOGIES



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Safety : Uday K, Ayush P	
Aerodynamics : G Y Rohit	
Chassis : Brian N Quadras	Workshop : Arunraj Subbaraj
Suspension : Kanal Sebastian	Admin : Arunraj Subbaraj
Brakes : Vamsi Naramsetty	Logistics : Pallav Rawat
Steering : Eshwarmurthy T	Finance : Mohammad Hannan
Engine : R Manjeet	SMP : Karan Chatrath
Drivetrain : Rajat Kabra	HR : Tejas Babaria
Electrical : Siva Subramanian	Testing : Rajit Malik
Coding : Arunraj Subbaraj	IT : Arunraj Subbaraj

STATIC EVENT MODULE COORDINATOR

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